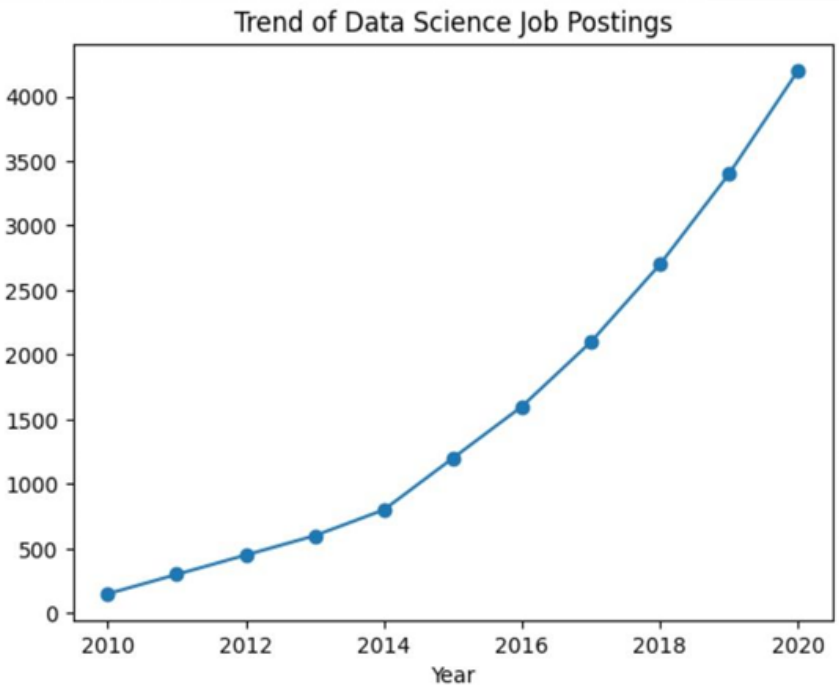
EXPERIMENT 1

1.a. Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
data ={'Year': list(range(2010,2021)),
'Job posting':[150,300,450,600,800,1200,1600,2100,2700,3400,4200]}
df=pd.DataFrame(data)
plt.plot(df['Year'],df['Job posting'],marker='o')
plt.title("Trend of Data Science Job Postings")
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

OUTPUT:

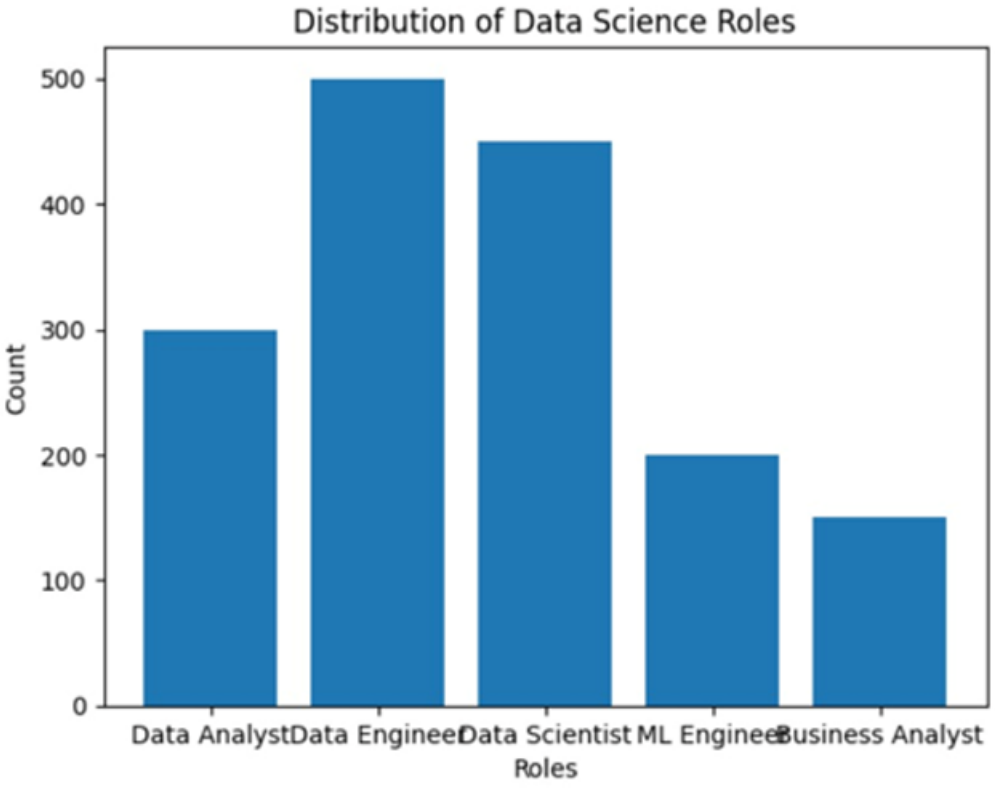


1.b.Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
roles=['Data Analyst','Data Engineer','Data Scientist','ML Engineer','Business Analyst']
counts=[300,500,450,200,150]
plt.bar(roles,counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Roles')
plt.ylabel('Count')
plt.show()
```

OUTPUT:



1.c.Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

PROGRAM:

```
import pandas as pd
structured_data=pd.DataFrame({
‘ID':[1,2,3],
 'Name':['Alice','Bob','Charlie'],
 'Age':[25,30,35]
})
 print("Structured Data:\n",structured_data)
unstructured_data="This is an example of unstructured data.it can be a piece of text, an image, or a video file."
print("\nUnstructured Data:\n",unstructured_data)
semi_structured_data={'ID':1,'Name':'Alice','Attributes':{'Height':165,'Weight':68}}
print("\nSemi-structured Data:\n",semi_structured_data)
```

OUTPUT:

```
Structured Data:
   ID     Name    Age
0   1     Alice     25
1   2     Bob       30
2   3    Charlie    35

Unstructured Data:
 This is an example of unstructured data.it can be a piece of text, an image, or a video file.
Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

1.d.Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

PROGRAM:

```
Import pandas as pd

from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Rajalakshmi Engineering College")

token

b'...'

f.decrypt(token)

b'Rajalakshmi Engineering College'

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABqfKxkWnnrfj9EG1MdQ8V5lo-
bltGUGZtSmxnPlf_5NqAxYxh3nRoOQcF_h-
nSVSpst6B6XnhQh2SehFrCdKAZrcm8MGcKc8sfA4R536lgLnjS47smFgZnHnAWJ8r9ZNKV
K4lb'

Decrypted Data: b'Rajalakshmi Engineering College.'
```